

COORDINATE GEOMETRY, TRIGONOMETRY AND NUMBER SYSTEM

Q1

Directions for questions 1 and 2: Type in your answer in the input box provided in the question

The distance of the point $(-5, -12)$ from the origin is

Solution:

Distance from origin to (x_1, y_1) is $\sqrt{x_1^2 + y_1^2}$

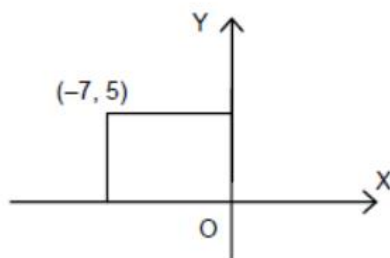
i.e. $\sqrt{(-5)^2 + (-12)^2} = 13$

Ans : (13)

Directions for questions 1 and 2: Type in your answer in the input box provided in the question

The perpendicular distance of y – axis from the point $(-7, 5)$ is

Solution:



Perpendicular distance from (x_1, y_1) to y – axis is $|x_1| = |-7| = 7$ units.

Ans : (7)

Directions for questions 3 to 7: Select the correct alternative from the given choices.

In which quadrant does the point of intersection of the lines $x = -4$ and $y = 7$ lie?

- ☐ a) Q_2
- ☐ b) Q_1
- ☐ c) Q_3
- ☐ d) Q_4

Directions for questions 3 to 7: Select the correct alternative from the given choices.

The ratio in which the y - axis divides the line segment joining the points $(-4, 3)$, $(-8, -4)$ is ____.

- ☐ a) 1 : 2 internally
- ☐ b) 2 : 1 internally
- ☐ c) 2 : 1 externally
- ☐ d) 1 : 2 externally

Directions for questions 3 to 7: Select the correct alternative from the given choices.

The slope of any line perpendicular to the line joining the points $(-4, 7)$ and $(5, 4)$ is ____.

- ☐ a) 3
- ☐ b) $\frac{1}{3}$
- ☐ c) $-\frac{1}{3}$
- ☐ d) -3

Solution:

Slope of the line joining the points (x_1, y_1) and (x_2, y_2) is $\frac{y_2 - y_1}{x_2 - x_1}$

Here $(-4, 7), (5, 4)$

$$\text{Slope of the line} = \frac{4 - 7}{5 - (-4)} = -\frac{1}{3}$$

Slope of the line perpendicular to the above line is $\frac{-1}{-1/3} = 3$

Choice (A)

Directions for questions 3 to 7: Select the correct alternative from the given choices.

The equation of the line joining the points $(-2, 3)$ and $(7, 2)$ is _____.

- ☐ a) $x - 9y + 25 = 0$
- ☐ b) $x + 9y - 25 = 0$
- ☐ c) $x - 9y - 25 = 0$
- ☐ d) $x + 9y + 25 = 0$

Solution:

Equation of the line joining the points (x_1, y_1) and (x_2, y_2) is $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$

$$\text{i.e. } y - 3 = \frac{2 - 3}{7 - (-2)} (x - (-2))$$

$$9(y - 3) = -(x + 2)$$

$$x + 9y - 25 = 0$$

Choice (B)

Directions for questions 3 to 7: Select the correct alternative from the given choices.

If $\sec \theta =$

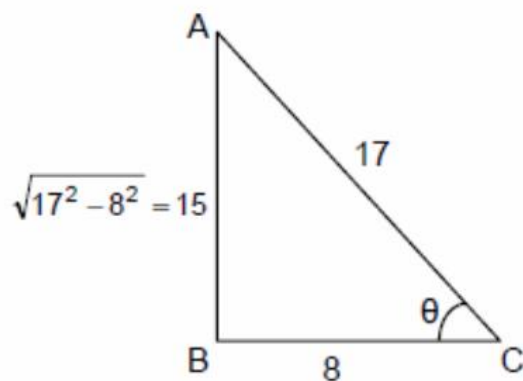
$$\frac{17}{8}$$

, $\theta \in Q_4$, then $\cos \theta - \sin \theta =$ ____.

- ☐ a) $\frac{23}{17}$
- ☐ b) $\frac{17}{23}$
- ☐ c) $-\frac{23}{17}$
- ☐ d) $-\frac{17}{23}$

Solution:

$$\sec \theta = \frac{17}{8} \quad \theta \in Q_4$$



$$\cos \theta = \frac{8}{17}$$

$$\sin \theta = \frac{-15}{17} \quad (\because \sin \theta \text{ is negative in } Q_4)$$

$$\begin{aligned} \cos \theta - \sin \theta &= \frac{8}{17} - \left(\frac{-15}{17} \right) \\ &= \frac{23}{17} \end{aligned}$$

Choice (A)

Directions for questions 8 and 9: Type in your answer in the input box provided in the question

If $3\cos^2\theta + 5\sin^2\theta = 4$, then $\tan \theta =$

$(0 < \theta < 90^\circ)$

Solution:

$$\begin{aligned}5\sin^2\theta + 3\cos^2\theta &= 4 \\2\sin^2\theta + 3(\sin^2\theta + \cos^2\theta) &= 4 \\2\sin^2\theta &= 4 - 3 \quad (\because \sin^2\theta + \cos^2\theta = 1) \\ \sin\theta &= \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ \\ \tan\theta &= \tan 45^\circ = 1\end{aligned}$$

Ans : (1)

Directions for questions 8 and 9: Type in your answer in the input box provided in the question

If $3\cos^2\theta + 5\sin^2\theta = 4$, then $\tan \theta =$
($0 < \theta < 90^\circ$)

Solution:

$$\begin{aligned}5\sin^2\theta + 3\cos^2\theta &= 4 \\2\sin^2\theta + 3(\sin^2\theta + \cos^2\theta) &= 4 \\2\sin^2\theta &= 4 - 3 \quad (\because \sin^2\theta + \cos^2\theta = 1) \\ \sin\theta &= \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ \\ \tan\theta &= \tan 45^\circ = 1\end{aligned}$$

Ans : (1)

Directions for questions 8 and 9: Type in your answer in the input box provided in the question

The value of $\operatorname{cosec}^2 70^\circ - \tan^2 20^\circ =$

Solution:

$$\begin{aligned} & \operatorname{cosec}^2 70^\circ - \tan^2 20^\circ \\ &= \operatorname{cosec}^2 (90^\circ - 20^\circ) - \tan^2 (20^\circ) \\ &= \sec^2 20^\circ - \tan^2 20^\circ \quad (\because \operatorname{cosec}(90 - \theta) = \sec \theta) \\ &= 1 \quad (\because \sec^2 \theta - \tan^2 \theta = 1) \end{aligned} \quad \text{Ans : (1)}$$

A wheel makes 210 revolutions in a minute. The angle made by the wheel in 4 seconds in radians is _____.

- ☐ a) 14π
- ☐ b) 28π
- ☐ c) 7π
- ☐ d) 24π

Solution:

$$\begin{aligned} & 1 \text{ mt} = 60 \text{ sec} = 210 \text{ revolutions} \\ & \text{In } 4 \text{ sec} = \frac{210 \times 4}{60} = 14 \text{ revolutions.} \\ & \text{Each revolution makes an angle of } 360^\circ \text{ or } 2 \pi \text{ radians} \\ & 14 \text{ revolutions make } 14 \times 2 \pi = 28 \pi \text{ radians} \end{aligned} \quad \text{Choice (B)}$$

Directions for questions 10 to 15: Select the correct alternative from the given choices.

If $\sec\theta + \tan\theta = 5$, then value of $\sin 2\theta$ is ____.

☐ a) $\frac{120}{169}$

☐ b) $\frac{26}{15}$

☐ c) $\frac{60}{13}$

☐ d) $\frac{125}{169}$

$$\sec\theta + \tan\theta = 5$$

$$\Rightarrow \sec\theta - \tan\theta = \frac{1}{5} \quad (\because \sec^2\theta - \tan^2\theta = 1)$$

by adding the above two equations

$$2\sec\theta = \frac{26}{5} \quad \sec\theta = \frac{13}{5}$$

$$\cos\theta = \frac{5}{13}$$

$$\sin\theta = \sqrt{1 - \cos^2\theta} = \sqrt{1 - \left(\frac{5}{13}\right)^2}$$

$$\sin\theta = \frac{12}{13}$$

$$\Rightarrow \sin 2\theta = 2\sin\theta \cos\theta$$

$$= 2 \times \frac{5}{13} \times \frac{12}{13}$$

$$= \frac{120}{169}$$

Choice (A)

Directions for questions 10 to 15: Select the correct alternative from the given choices.

The value of $\cot \theta$ when expressed in terms of $\sec \theta$ is

☐ a) $\frac{\sec \theta}{\sqrt{1 - \sec^2 \theta}}$

☐ b) $\frac{1}{\sqrt{1 + \sec^2 \theta}}$

☐ c) $\frac{1}{\sqrt{\sec^2 \theta - 1}}$

☐ d) $\frac{2 \sec \theta}{\sqrt{1 + \sec^2 \theta}}$

Solution:

$$\begin{aligned}\cot \theta &= \frac{\cos \theta}{\sin \theta} = \frac{1/\sec \theta}{\sqrt{1 - \cos^2 \theta}} \\ &= \frac{1/\sec \theta}{\sqrt{1 - 1/\sec^2 \theta}} = \frac{1/\sec \theta}{\frac{\sqrt{\sec^2 \theta - 1}}{\sec \theta}} \\ &= \frac{1}{\sqrt{\sec^2 \theta - 1}}\end{aligned}$$

Choice (C)

Directions for questions 10 to 15: Select the correct alternative from the given choices.

The value of $(101101)_2 + (10111)_2$ is _____.

- ☐ a) $(100100)_2$
- ☐ b) 110011
- ☐ c) $(1001100)_2$
- ☐ d) $(1000100)_2$

Solution:

16		43852	
16		2740	--- C
16		171	--- 4
		A	--- B



$$\therefore (43852)_{10} = (AB4C)_{16}$$

Choice (B)

Solution:

$$\begin{array}{r} 101101_2 \\ + 10111_2 \\ \hline 1000100_2 \end{array}$$

$$\therefore (101101)_2 + (10111)_2 = (1000100)_2$$

Choice (D)

Directions for questions 10 to 15: Select the correct alternative from the given choices.

The hexa decimal equivalent of $(43852)_{10}$ is ____.

- ☐ a) $(AC4B)_{16}$
- ☐ b) $(AB4C)_{16}$
- ☐ c) $(4ABC)_{16}$
- ☐ d) $(ABC4)_{16}$

The octal equivalent of the binary number 101110111 is _____.

- ☐ a) $(567)_8$
- ☐ b) $(456)_8$
- ☐ c) $(756)_8$
- ☐ d) $(657)_8$

Solution:

$(101110111)_2$

For converting to octal base, we group 3 digits from right to left as shown above and convert each group of three digits to decimal and write.

$$111 = 1 \times 2^2 + 1 \times 2 + 1 \times 2^0 = 7$$

$$110 = 1 \times 2^2 + 1 \times 2 + 1 \times 0 = 6$$

$$101 = 1 \times 2^2 + 0 \times 2 + 1 \times 2^0 = 5$$

$$\therefore (101110111)_2 = (567)_8$$

Choice (A)